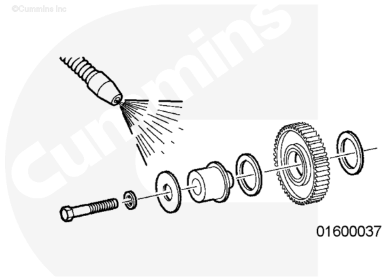
Trainings

Clean and Inspect for Reuse

The shaft for the water pump idler gears and camshaft idler gears are **different**. The water pump idler shaft has an integral flange which the camshaft idler shafts do **not**.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

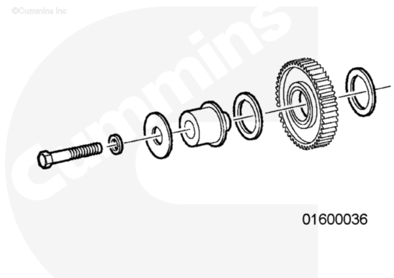
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent. Clean the parts.

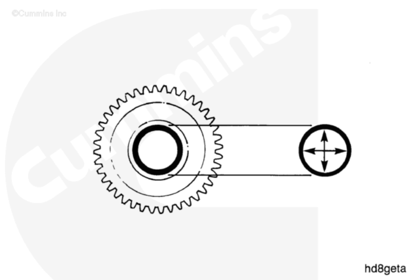
Dry with compressed air.

Check the parts for damage.



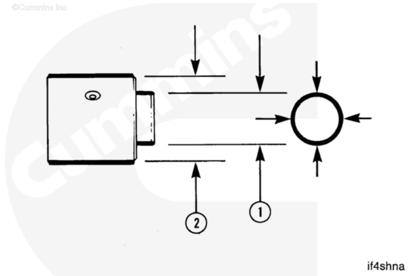
Measure the inside diameter.

Idler Gear Bushing Inside Diameter		
mm		in
47.638	MIN	1.8755
47.714	MAX	1.8785



Note : The bushing in the gear is precision bored after installation. If machining capability is not available, replace the bushing and the gear as an assembly.

Note : Some engines have a camshaft idler shaft that is the same as the water pump idler shaft. Refer to Procedure 001-999 in Section F. (/qs3/pubsys2/xml/en/procedures/28/28-001-999-tr.html)



Measure the outside diameter.

Water Pump Idler Shaft Outside Diameter — Engine Serial Number Higher than 33110701 (1)		
mm		in
43.167	MIN	1.6995
43.180	MAX	1.7000

Water Pump Idler Shaft Outside Diameter — Engine Serial Number Higher than 33110701 (2)		
mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Water Pump Idler Shaft Outside Diameter — Engine Serial Number Lower than 33110701 (1)		
mm		in
23.397	MIN	0.9995
25.400	MAX	1.0000

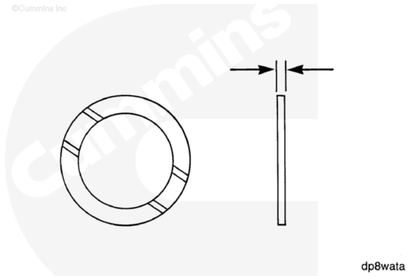
Water Pump Idler Shaft Outside Diameter — Engine Serial Number Lower than 33110701 (2)		
mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Note : Some engines with an Engine Serial Number less than 33110701 have the shaft with the larger diameter (1). These engines were modified in the field using a service tool that is available to modify the cylinder block.

Check the grooved side for damage.

Measure the thickness.

Thrust Bearing Thickness		
mm		in
2.34	MIN	0.092
2.39	MAX	0.094



Note : Oversize thrust bearings are available to adjust the end clearance.

Last Modified: 24-Sep-2015